Speaker script

Measuring Agglomeration — Homework 2, Spatial & Regional Economics

Full speech for all 20 slides, one speaker. About 24–25 minutes at a calm pace: Part A about 8 minutes, Part B about 7, Part C about 7, plus introduction and closing. If only one part is drawn, use slides 1–2, that part, and slide 20.

How to use this script.

- The **blue-bordered text** is what you say. It is written to be spoken: short sentences, one idea each.
- // marks a natural pause: look at the slide, let the number sink in.
- Formulas are written as you would *say* them; the slide shows them in symbols.
- **Numbers to remember** lists the figures on that slide. **If asked** gives a short answer to the most likely question.
- The Italian line at the end of each slide is the idea in one sentence, to check you understood it, not to say aloud.
- When a slide shows code, explain *what the block does*, not every character: say which act it follows and which variable carries the answer.

Slide 1 — Measuring Agglomeration

INTRODUCTION · 20 s

> Good morning, Professor. We are Davide Maiellaro, Caterina Congia, Federica Stella Favalli and Lorenzo Piva.
>
> Homework 2 is about one parameter, the productivity spillover **lambda**: how much a location's productivity rises when more people work there.

Slide 2 — One parameter, three questions

INTRODUCTION · THE COMMON THREAD · 1 min

> The whole homework turns around one equation: A equals A-bar times L to the lambda. A-bar is what a place intrinsically is — its fundamental productivity. L to the lambda is what it gains from its own size.
>
> Why does it matter? // Because lambda decides how much a local gain — a new plant, a subsidy, an infrastructure — is *amplified* by the workers it attracts. And it decides whether there is an efficiency case for subsidising density at all.
>
> The problem is measurement. In **Part A** we show that a cross-section fits every value of lambda equally well. So the answer must come from shocks that move people while fundamentals stay put. **Part B** uses the division of Germany, a shock to market access *across* cities. **Part C** uses the Berlin Wall, a shock *within* a city, to estimate lambda and to evaluate a metro line.

In una frase: tutto ruota attorno a λ: conta per le politiche, ma una sezione trasversale non lo identifica; servono shock (Germania divisa, Muro di Berlino).

Slide 3 — Productivity now depends on size

PART A · POINT (A) · 1 min 30 s

> Part A, point (a). We take the Armington model of Topic 6 and add a production spillover: A-i equals A-bar-i times L-i to the lambda. Nothing else changes.
>
> Each location produces one variety with labour only. So the price of i's good delivered to n is tau-n-i times w-i, divided by A-bar-i L-i to the lambda.
>
> Three equations. // The **price index**: P-n to the one minus sigma is the sum over suppliers of phi-n-i times w-i over A-bar-i L-i to the lambda, all to the one minus sigma. The **trade shares**: each supplier's term divided by the price index — the gravity equation, cheaper and closer suppliers sell more. The **trade balance**: the wage bill of i equals its sales to all destinations, w-i L-i equals the sum over n of pi-n-i w-n L-n. Rents are redistributed to residents and spent on land, so spending on goods is w-n L-n.
>
> The key observation is in the box at the bottom. // A-bar and lambda *never appear separately*: only in the product A-bar L to the lambda, which is A. Rents, utility and the Fréchet mobility shares contain no productivity. So two parametrisations with the same A in every location are the **same economy**.

Numbers to remember $A_i = \bar{A}_i L_i^{\lambda}$ · $X_n = w_n L_n$ (rents redistributed) · $\bar{A}$ and λ enter only through A.

If asked: "Where does L appear in the trade balance?" — Twice: as the labour force, $w_i L_i$, and inside productivity, with exponent $\lambda(\sigma - 1)$ in the trade share. That is what makes the spillover an agglomeration force.

In una frase: con lo spillover cambiano solo le produttività; $\bar{A}$ e λ compaiono sempre insieme nel prodotto A.

Slide 4 — The data identify A, not its split

PART A · POINT (A), IDENTIFICATION · 1 min 30 s

> What does the inversion recover? Given population, wages, the trade costs and sigma, the trade balance is N equations in N unknowns, the ratios w over A. A fixed point solves it, and gives A **exactly**, up to a common scale. // This step uses L and w as data and *never uses lambda*. In the code it is `inv.Aeff`.
>
> To get the fundamental we need lambda: A-bar equals A times L to the minus lambda. That is `inv.A`, and it changes with lambda.
>
> And every lambda fits equally well. Take any lambda-prime and define A-bar of lambda-prime as A times L to the minus lambda-prime. Then A-bar times L to the lambda-prime is A again, in every location. Prices, shares, the trade balance: all identical, and the data

are reproduced exactly. // N equations pin down N composites — not N fundamentals *plus* lambda.

So: is a city productive because of *where* it is, or because it is *big*? One cross-section cannot tell. It is the same logic as Part C of Homework 1.

But lambda does not change the fit, and it does change every counterfactual: after a shock workers move, and with lambda positive productivity moves with them. That is point (c).

If asked: "Why up to a scale?" — Trade shares depend only on ratios of w/A; multiplying all A by a constant changes nothing real. The code fixes it with a geometric-mean normalisation.

In una frase: l'inversione dà A esatto senza usare λ; per $\bar{A}$ serve λ, e ogni λ riproduce i dati.

Slide 5 — Same data, three values of lambda

PART A · POINT (B) · 1 min 30 s

Point (b), in numbers. In ARM1 we added a section after act 6. The data are the baseline equilibrium of act 5, generated with lambda equal to zero.

Look at the code. We loop over three values, zero, 0.07 and 0.25. The important line is the comment: // lambda is changed *only in a copy* of the parameters, p equals par. If we changed it at the top of the script, we would change the data themselves. Then we call `arm_invert` with p, and record two things: the error of `inv.Aeff` against the true productivity, and the standard deviation of log A-bar.

The table. The error is **1.17 times ten to the minus twelve** in all three rows — machine precision — and even the number of iterations is the same, 4346. Why? Because lambda enters `arm_invert` only *after* the loop, in A equals Aeff divided by L to the lambda. Same composite, same error. That is point (a) in numbers.

The fundamentals instead change: the dispersion of log A-bar falls from **0.303 to 0.272 to 0.204**. // The formula on the right explains it: the variance of log A-bar is the variance of log A, minus two lambda times the covariance with log L, plus lambda squared times the variance of log L. Productive places are big — the correlation is 0.814 — so a larger lambda credits more productivity to size, and fundamentals look more alike. The variance keeps falling as long as lambda is below 0.43, the OLS slope of log A on log L.

Same data, same fit, different story about where productivity comes from.

Numbers to remember Error 1.17×10^{-12} for all λ · 4346 iterations · $\mathrm{sd}(\log \bar{A})$: 0.3033, 0.2715, 0.2041 · $\mathrm{corr}(\log A, \log L) = 0.814$ · turning point $\lambda = 0.43$.

If asked: "Does the formula really reproduce the table?" — Yes: with $\mathrm{sd}(\log A) = 0.3033$, $\mathrm{sd}(\log L) = 0.575$ and correlation 0.814 it gives 0.2715 and 0.2041.

In una frase: stesso errore per ogni λ perché λ entra solo dopo il punto fisso; la dispersione di $\bar{A}$ cala perché più λ attribuisce produttività alla dimensione.

Slide 6 — The same shock attracts far more people

PART A · POINT (C) · 1 min 15 s

> Point (c). We repeat act 5: the same fundamentals and the same **plus 20 percent** shock to A-bar in location 8, the middle of the line, for the three values of lambda.
>
> In the code the key is that p goes into *both* calls to `arm_solve`: the baseline and the shocked economy. The baseline also changes with lambda, because A equals A-bar L to the lambda.
>
> The table. // The population of the shocked location grows by **26.9 percent** with lambda zero, **30.3** with 0.07, and **45.3** with 0.25. An amplification of 1.13 and 1.68. With lambda zero we recover act 5 exactly.
>
> And there is an exact check, the formula on the right: one plus the change in A-j equals 1.20 times one plus the change in L-j, to the lambda. At 0.25: 1.20 times 1.4528 to the 0.25 is 1.317 — the plus 31.7 in the table. // So the shock to the fundamental is always 20 percent, but the productivity the economy actually sees is not.

Numbers to remember ΔL_8: +26.91%, +30.34%, +45.28% · ratio 1.00, 1.13, 1.68 · ΔA_8: +20.00%, +22.25%, +31.74%.

In una frase: lo stesso shock del 20% attira il 27% di persone in più senza spillover e il 45% con $\lambda = 0.25$.

Slide 7 — Workers bring productivity with them

PART A · POINT (C), MECHANISM · 1 min 15 s

> Why does the response grow with lambda? Follow the loop on the right.
>
> The real wage rises in j, so workers migrate there — that is the **labour market**. With lambda positive they bring productivity with them: A-j rises. Then the **goods market**: higher productivity makes j's good cheaper everywhere, its sales and wages rise, utility rises further, and more workers come.
>
> The neighbours face two forces. // Migration pulls workers toward j; but they buy a lot of j's good, which is now cheaper, so their price index falls. With lambda positive the second force is stronger: the right neighbour goes from **minus 0.28** to **plus 2.46 percent**. Far from j only the migration pull remains: about minus 1.8 percent at lambda zero, minus 2.4 at 0.25.
>
> So the spillover makes the shock larger and more concentrated in space. // And since all three lambdas fit the same data, the policy answer — 27 or 45 percent — depends on a parameter the cross-section cannot identify.

If asked: "Why do the two neighbours differ?" — Their own fundamentals are drawn at random, so they start from different sizes and trade shares with j.

In una frase: i lavoratori portano produttività; i vicini perdono per la migrazione ma guadagnano dal bene più economico di j.

Slide 8 — Helpman in disguise

PART A · POINT (D) · 1 min 15 s

> Point (d), Helpman 1998, in steps.
>
> In Helpman each location produces many varieties. Free entry makes their number proportional to population, n-i equals kappa L-i, and each sells at a constant markup over marginal cost.
>
> Step one: the contribution of i to the price index of n is n-i times the price to the one minus sigma — that is **L-i to the power one**, times a term without L. Step two: in Armington with the spillover the contribution is **L-i to the lambda times sigma minus one**, times a term without L.
>
> Step three: // the two coincide when lambda times sigma minus one equals one. So **lambda equals one over sigma minus one**: 0.25 at sigma 5, the value in ARM1; one third at sigma 4, the value in ARM2 and in Redding and Sturm. Keep this one third in mind for Part B.
>
> Step four: an estimated lambda need not be a technological externality, like knowledge spillovers. It can be the *variety gains* of monopolistic competition: more people, more firms, more varieties, a lower price index. Same equations, different story.

Numbers to remember $\lambda(\sigma-1)=1 \Rightarrow \lambda = 1/(\sigma-1)$ · $\sigma=5$: 0.25 · $\sigma=4$: 1/3.

In una frase: un $\lambda = 1/(\sigma-1)$ è equivalente a Helpman: l'effetto può essere varietà, non tecnologia.

Slide 9 — What would identify lambda?

PART A · POINT (E) · 1 min

> Point (e). If one cross-section cannot identify lambda, because the fundamentals adjust to rationalise any value, we need **variation over time**: a panel.
>
> What must change? Population, for reasons unrelated to productivity: a border that closes, a country divided, a new infrastructure. What must stay fixed? The fundamentals: the shock must move people without changing what each place intrinsically is.
>
> Then we invert the model at both dates. // With A-bar fixed, the change in log A is lambda times the change in log L. That relation identifies lambda.
>
> And this is exactly what the next two parts do.

In una frase: serve uno shock che sposti la popolazione lasciando fermi i fondamentali; invertendo prima e dopo si stima λ.

Slide 10 — A fake border finds nothing

PART B · POINT (A) · 1 min 30 s

Part B: Redding and Sturm. After the division of Germany, West German cities within 75 kilometres of the new border lost population relative to the rest of the West. The figure is their design: treatment and control have the same trend until 1949, and diverge after.

Point (a) is a placebo. We draw a **fake border** on the meridian 7.5 degrees East. In the code: `dmer` is each city's distance from the meridian in kilometres; `samp` is the West without the 20 real border cities; `pl_t` and `pl_c` are the placebo treated — the 58 cities within 75 km, mostly the Rhine–Ruhr — and the 41 controls; `did_pl` is the difference-in-differences of annual growth, post-war minus pre-war, as in act 0.

The result. // The placebo is **minus 0.064** percentage points per year, t equal to minus 0.28, p equal to 0.78. The actual border gives **minus 0.688**, t equal to minus 3.9 — or minus 0.75 in Redding and Sturm's specification. The placebo is less than one tenth of the real effect, and statistically zero.

Numbers to remember Placebo -0.064 pp/yr ($t = -0.28$, $p = 0.78$) · actual -0.688 ($t = -3.90$); RS -0.75 · 58 treated, 41 control, 194 km from the border.

If asked: "Why drop the 20 border cities?" — Otherwise the real treatment contaminates the placebo control group.

In una frase: un confine finto sul meridiano 7.5°E non trova nulla: l'effetto è specifico del confine vero.

Slide 11 — The null placebo supports the market-access reading

PART B · POINT (A), INTERPRETATION · 1 min

What would a large negative placebo have meant? Redding and Sturm's argument is that the border is an exogenous shock to market access: without it, border cities and the rest of the West would have developed alike.

A large negative estimate along an arbitrary line would say that western regions were *already* on different paths, or hit by other shocks — bombing, the decline of the Ruhr, where refugees settled. // Parallel trends would fail, and the minus 0.75 could not be read as the effect of lost market access. Our placebo is zero, so the loss is specific to the border.

The optional part: the same placebo *in the model*. Using the simulated division of act 2, `mod_pa`, for the same groups, the model gives **plus 0.052**. Why positive? The meridian strip is the part of the West farthest from the border: its exposure to the East is lower, 0.085 against 0.122, so it loses less and grows relative to the rest. The size is about half of the model's own treatment effect, minus 0.112, with the opposite sign.

In una frase: un placebo grande avrebbe smentito i trend paralleli; nel modello il placebo è leggermente positivo perché la fascia del meridiano è lontana dall'Est.

Slide 12 — Sealing the border deletes the eastern terms

PART B · POINT (B), ALGEBRA · 1 min 15 s

Point (b): exposure. The price index is the sum over suppliers of phi-n-i times w-i over A-i, to the one minus sigma. By the definition of the trade share, each term is pi-n-i times the price index. So the eastern terms add up to **s-E-n** times the price index, where s-E-n is the share of n's spending on eastern cities.

Sealing the border sets phi to zero for those suppliers. At given wages, the new price index to the one minus sigma is just the western terms: one minus s-E times the old one. // So the price index rises by the factor one minus s-E, to the minus one over sigma minus one.

The bound. Shares sum to one, and the western shares include the home share. So s-E equals one minus the western shares, which is at most **one minus pi-n-n**. Equality only if n buys nothing from other western cities.

The intuition: a city that buys mostly at home cannot be very exposed. A city that buys a lot elsewhere *can* be — if that elsewhere is the East.

Numbers to remember $P_n'/P_n = (1 - s_n^E)^{-1/(\sigma-1)}$ · $s_n^E \leq 1 - \pi_{nn}$.

In una frase: chiudere il confine elimina i termini orientali dell'indice dei prezzi; l'esposizione è al massimo uno meno la quota interna.

Slide 13 — Exposure ranks the losers almost perfectly

PART B · POINT (B), IN THE MODEL · 1 min 15 s

Now in the model, at the 1939 equilibrium. The code: `sE` sums the trade shares of each western city over the eastern columns; `pnn` is the home share; then we correlate both with the simulated population change.

The bound holds in all 119 West cities. Mean exposure is 0.113: **0.176** for the treated border cities, 0.100 for the control. At given wages the price index rises by 6.5 percent in treated cities against 3.5.

The left panel: // the correlation between exposure and the simulated change is **minus 0.998**. Exposure ranks the losers almost perfectly; general equilibrium rescales the shock but keeps the order.

The home share works only among the treated: plus 0.854, because those 20 cities are at a similar distance from the border, so a low home share means trading with the East. Across all 119 it drops to 0.213: in the Rhineland a low home share means trade with other western cities, which the border does not touch.

> The link to act 3: small cities buy less at home, 0.37 against 0.54, so they are more exposed and lose more: minus 0.140 against minus 0.067 in the model, minus 1.12 against minus 0.37 in the data.

Numbers to remember $\text{corr}(s^E, \Delta) = -0.998$ · $\text{corr}(\pi_{nn}, \Delta)$: +0.854 (20 treated), +0.213 (119) · mean s^E: 0.176 vs 0.100 · corr with data −0.079.

In una frase: nel modello l'esposizione all'Est ordina quasi perfettamente chi perde; la quota interna funziona solo tra le città di confine.

Slide 14 — The dials change the size, not the ranking

PART B · POINT (C) · 1 min 15 s

> Point (c), optional. At the baseline, the first row, the model gives **minus 0.11** percentage points per year against **minus 0.75** in the data, and a correlation of only **0.08** across cities. The first number is about the *size*; the second about the *ranking*.
>
> For the size, the model can accommodate a stronger response: more mobility, epsilon; more agglomeration, lambda; steeper distance decay; lower sigma. It cannot accommodate channels that are not market access: border cities losing their role as hubs or administrative centres, firms moving west, war damage and refugees.
>
> For the ranking, it can accommodate unequal fundamentals and real transport networks, but not census noise and city-specific shocks: the cross-city spread of growth is 1.09 in the data against 0.07 in the model.
>
> The highlighted row is act 5: // epsilon 100 and lambda one third. The effect goes from minus 0.11 to **minus 0.34**, but the correlation with the data stays at **0.079**. The row below is Redding and Sturm's own simulation, which has epsilon *equal to infinity* and lambda one third. The parameters scale the response; they do not change who loses.

Numbers to remember Baseline −0.112, corr 0.076 · act 5 ($\varepsilon = 100$, $\lambda = 1/3$) −0.342, corr 0.079 · RS ($\varepsilon = \infty$) −0.362 · data −0.746; sd 1.09 vs 0.07.

In una frase: i parametri cambiano la dimensione dell'effetto ma non l'ordinamento delle città, che resta poco correlato con i dati.

Slide 15 — Two models agreeing, not a model fitting the data

PART B · POINT (D) · 1 min 15 s

> Point (d). In act 5, epsilon 100 and lambda one third give **minus 0.34**, against **minus 0.36** in Redding and Sturm's simulation. Which assumption on agglomeration is implicit in their model?
>
> Go back to point (d) of Part A. // Helpman's free entry makes varieties grow with city size: an agglomeration force of exactly one over sigma minus one — one third at sigma 4. That is why no lambda appears in their paper: it is not a free parameter, it is fixed by monopolistic competition and sigma. Our model makes it an explicit dial.
>
> The code recalibrates and re-solves the division with those parameters, `p3`, and correlates the result with the data and with their simulation. City by city, the correlation with Redding and Sturm is **0.99997**. Raising epsilon to 300 and 1000 gives minus 0.354 and minus 0.359, converging to their minus 0.362.
>
> So is this evidence about the data? // No: it is evidence that the two models are *the same system*. Both under-predict the data, minus 0.36 against minus 0.75, and both have the same low correlation with city growth, about 0.08.

If asked: "Why does ε have to be large?" — Redding and Sturm assume perfect mobility, utility equalised everywhere; in the Fréchet model that is the limit $\varepsilon \to \infty$.

In una frase: RS assume implicitamente $\lambda = 1/3$ (Helpman); l'accordo tra i due modelli dice che sono lo stesso sistema, non che spiegano i dati.

Slide 16 — OLS finds three times the true lambda, even where there is none

PART C · POINT (A) · 1 min 30 s

> Part C: the Berlin Wall, Ahlfeldt, Redding, Sturm and Wolf. Point (a). We ran ARSW0 as it is. It simulates a city where the true lambda is **0.07**, and regresses log productivity on log surrounding employment density.
>
> The table. With random fundamentals, OLS gives **0.206** on average over 20 draws. When fundamentals favour the centre, **0.218**. // Both are about *three times* the true value. In both panels the red OLS line is much steeper than the dashed true slope.
>
> Why? The formula below the table. Log productivity is log A-bar plus lambda log Upsilon, so the fundamental sits in the error term, and it is correlated with density. In case (a) fundamentals are unrelated to location, so the bias is pure **reverse causality**: blocks with a high fundamental attract workers, which raises the density around them. Density responds to productivity instead of causing it. In case (b) there is also an **omitted variable**: the centre is both denser and intrinsically better. // But it adds only about 0.012: reverse causality drives most of the bias.

Policy. A government doubling density expects two to the 0.07 minus one, about **5 percent**, with the true value — but **15.35 percent** with OLS, three times as much. // And in case (c), a city with *no* spillover at all, OLS still reports 0.196, almost the same number. The government would expect a 14.6 percent gain and subsidise concentration where there is no market failure to correct.

Numbers to remember True $\lambda = 0.07$ · OLS: 0.206 (0.177–0.247), 0.218 (0.206–0.225), 0.196 when $\lambda = 0$ · doubling density: 4.97% vs 15.35% (14.6% in the $\lambda = 0$ city).

If asked: "Why is there a bias in case (a), if fundamentals are random?" — Random in *space*, not unrelated to density: a block with a good draw attracts workers, so the density around it rises. The covariance between density and the fundamental is positive even without any geography.

In una frase: l'OLS trova circa tre volte il vero λ perché la densità risponde alla produttività; trova 0.20 anche in una città senza spillover.

Slide 17 — The division of the city recovers the true lambda

PART C · POINT (B) · 1 min 30 s

Point (b): act 5 of ARSW2, the professor's updated script, run as it is. The estimator uses the division of the city. For each candidate lambda, it inverts the model before and after the Wall, recovers the fundamentals, and computes the gradient of their change in distance to the old centre, in the West. // The estimate is the lambda at which that gradient is **zero**.

The figure. With fundamentals unchanged, the blue line crosses zero exactly at **0.0700**, the true value. With random drift in fundamentals, five draws give 0.053, 0.070, 0.075, 0.071 and 0.087: mean **0.071**, range 0.034. Five draws are a suggestive check, not a formal proof, but they scatter around the true value.

The identifying assumption: // the Wall changed what blocks could *reach*, not what they *are*. The change in fundamentals is uncorrelated with distance from the old centre. Then any systematic change in recovered productivity with distance must come from the loss of surrounding density.

The moment at lambda zero is **plus 0.0204**: if we assume away agglomeration, the model reads the lost spillovers as a fall in fundamental productivity near the centre, for no reason.

Why do city governments need this and not OLS? Because OLS gives about 0.20 *whether the true lambda is 0.07 or zero*, while this estimator uses an exogenous shock that holds fundamentals fixed.

Numbers to remember $\hat{\lambda} = 0.0700$ · with drift: 0.053, 0.070, 0.075, 0.071, 0.087 (mean 0.071, range 0.034) · moment at $\lambda = 0$: +0.0204.

If asked: "When would the assumption fail?" — If something else hit the old centre at the same time as the Wall — say, war damage or a policy concentrated there — the fundamentals would change with distance and the estimate would be biased.

In una frase: con il Muro si sceglie il λ per cui i fondamentali non cambiano con la distanza dal centro: si ritrova 0.07.

Slide 18 — A wrong lambda overstates the welfare gain by 56%

PART C · POINT (C) · 1 min 45 s

Point (c). Act 6 of ARSW2 builds a metro line along the central row of the city: trains at 3 minutes per kilometre, 2 minutes of waiting, a walk to the station in your own column, and every commuting pair takes the faster of walking and the metro.

Our only change to the code is `lambda_policy`: we ran the act three times, with zero, 0.07 and 0.18. For each value the model recovers the fundamentals from the *same* observed city, so before the line the three cities are identical — the check reproduces the data with zero error. Only the prediction after the line differs. One detail, in the footnote: at lambda zero only the *production* spillover is removed; the residential externality, eta 0.15, stays on.

In the table, blue is an increase and red a decrease.

(i) Who gains. 6,476 of the 20,449 commuting pairs get faster, about a third, saving a median of 4.4 minutes. Commuters along the corridor and owners of floor space near the line gain: in spatial equilibrium the saving is *capitalised* in floor prices, which rise on the line, by 13 to 32 percent, and fall away from it.

(ii) Central jobs. // With lambda zero, the metro only makes commuting cheaper: workers can live farther away, nothing anchors firms in the centre, and central jobs *fall* by 3.85 percent. With lambda 0.07, more workers reach the centre, density rises, and density raises productivity there: central jobs *grow* by 2.49 percent. The sign flips.

(iii) The wrong lambda. A government using the OLS value, 0.18, would predict a welfare gain of **2.60 percent** instead of **1.67**. // That is 1.56 times: a 56 percent overstatement, 0.93 percentage points. It would also predict an implausible jump of 52 percent in central jobs and a fall of 35 percent in residents near the line. And since the updated OLS estimate is actually 0.206, using 0.18 is a *conservative* measure of the distortion.

Numbers to remember Welfare +1.37 / +1.67 / +2.60% · central jobs −3.85 / +2.49 / +52.08% · prices on the line +13.04 / +15.57 / +32.30% · residents near the line +16.21 / +9.54 / −35.08% · ratio $2.60/1.67 \approx 1.56$.

If asked: "Why do residents near the line fall by 35% with $\lambda = 0.18$?" — The agglomeration feedback is so strong that firms crowd into the centre: commercial demand bids up floor prices along the line and, with fixed floor space, residents are pushed out.

In una frase: la metro avvantaggia chi vive e possiede spazio lungo la linea; con il λ dell'OLS il guadagno di benessere previsto è sovrastimato del 56%.

Slide 19 — 0.07 is a block-level number

PART C · POINT (D) · 1 min 15 s

> Point (d), optional. Is 0.07 a good value for `par.lambda` in a regional model like ARM2?
>
> Start from the banner. In ARSW the spillover decays with travel time, delta 0.36 per minute: it **halves every 1.9 minutes**, about 160 metres on foot. So it is negligible beyond a few hundred metres. ARM2 links cities hundreds of kilometres apart.
>
> **For:** 0.07 is a standard order of magnitude, and without regional estimates it gives an upper-bound scenario for policy. Some mechanisms — thick labour markets, shared input suppliers — reach beyond the city.
>
> **Against**, four arguments. **Decay**: lambda is inseparable from its decay rate. **Mechanisms**: blocks interact face to face, regions through freight and trade. **The model**: as we saw in Part B(d), in ARM2 a lambda of one over sigma minus one, 0.25 to 0.33, is a Helpman variety gain, a different object. **Evidence**: regional estimates, such as Rosenthal and Strange 2004 and Combes and Gobillon 2015, are lower, 0.02 to 0.05.
>
> What would we need? // A decay function for inter-regional distances; lambda estimated on regional data, NUTS-2 or NUTS-3; and measured trade costs, to separate spillovers from the market access that ARM2 already captures.
>
> Our conclusion: at the regional scale, 0.07 is at most an **upper bound**.

Numbers to remember $\delta = 0.36/\text{min}$ · half-life $\ln 2/0.36 \approx 1.9$ min ≈ 160 m · regional estimates 0.02–0.05 · ARM2: $1/(\sigma - 1) \approx 0.25$–$0.33$.

In una frase: 0.07 misura spillover tra isolati a poche centinaia di metri; per un modello regionale è al massimo un limite superiore.

Slide 20 — Takeaways

CLOSING · 1 min

> To sum up.
>
> **First**, Part A: a cross-section identifies A, not lambda. The model recovers productivity exactly, but not how much comes from density. And lambda matters: the same shock moves 26.9 percent more people without spillovers, 45.3 percent with lambda 0.25.
>
> **Second**, Part B: shocks to market access move people. Exposure to the East ranks the losers of the division; mobility and agglomeration change the size of the effect, not the ranking. And Redding and Sturm's implicit lambda of one third is Helpman's one over sigma minus one.

> **Third**, Part C: // lambda needs a natural experiment. OLS gives about 0.21 for a true 0.07, and 0.20 even when there is no spillover; the Wall recovers 0.07; and with the OLS value the welfare gain of a metro is overstated by 56 percent.
>
> Thank you for your attention. I am happy to take your questions.

In una frase: tre messaggi: la sezione trasversale non identifica λ; gli shock di accesso al mercato spostano persone; per stimare λ serve un esperimento naturale.

Appendix — likely questions and short answers

- **What did you change in the code?** Part A: a section after act 6 of ARM1 (inversion with three λ) and act 5 repeated with `p` in place of `par`. Part B: blocks in ARM2 after acts 0, 2, 3 and 5 (placebo, model placebo, exposure, Helpman comparison). Part C: only `lambda_policy` in act 6 of ARSW2, the professor's updated script; ARSW0 and act 5 were run as given.
- **Why change λ only in a copy of `par`?** Changing it at the top would regenerate the data with a different λ; we want the *same* data inverted with different assumptions.
- **Why is λ not identified, but A is?** N equations (trade balance) for N unknowns w_i/A_i. Splitting each A_i into $\bar{A}_i$ and a common λ adds one unknown with no new equation.
- **Is the spillover a technology or a variety effect?** The model cannot say: with $\lambda = 1/(\sigma - 1)$ Armington and Helpman give identical price indices.
- **Why is the placebo positive in the model?** The meridian strip is far from the border, so less exposed (s^E 0.085 vs 0.122): it loses less than the control.
- **Why does the home share fail across all 119 cities?** A low π_{nn} can mean trade with the East (exposed) or with other western cities (not exposed); only among border cities does it mean the first.
- **Why does the model correlate so little with the data (0.08)?** The data contain city-specific shocks and noise (sd 1.09 pp/yr vs 0.07 in the model); the model has only market access.
- **What is the bias in naive OLS?** $\operatorname{plim} \hat{\lambda} = \lambda + \operatorname{cov}(\log \Upsilon, \log a)/\operatorname{var}(\log \Upsilon)$: positive because density follows fundamentals (reverse causality, most of the bias) and the centre is better (omitted variable, only +0.012).
- **Why does the Wall-based estimator work with drifting fundamentals?** The drift is random, unrelated to distance from the centre, so it adds noise but no bias: mean 0.071 over five draws.
- **Why do central jobs fall with $\lambda = 0$?** Cheaper commuting lets workers live farther away; without a spillover nothing keeps firms concentrated.
- **Where does 0.02–0.05 come from?** Estimates at the regional or metropolitan level, e.g. Rosenthal and Strange (2004), Combes and Gobillon (2015); we use them only to argue that 0.07 is an upper bound at that scale.